

NON-MF GROUPS EXIST: A CONCRETE FINITELY PRESENTED COUNTEREXAMPLE

ANONYMOUS

ABSTRACT. A countable discrete group is MF here if it embeds in the unitary group of a norm matrix corona. We construct an explicit finitely presented group that is not MF. The mechanism is a rigidity theorem for one-sided compressions of Kazhdan subgroups: in operator-norm matrix models, every compressor preserves the asymptotic commutant. Thus a certain commutator u converges to 1 in normalized Hilbert–Schmidt norm. A Clifford representation makes the central sign u^2 equal to -1 , giving the contradiction.

More generally, every finitely presented property-(T) group with a proper injective self-embedding produces a finitely presented non-MF group. We give an eight-generator instance based on $\mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z})$. Consequently MF groups are not closed under quotients, and the reduced group C^* -algebra of this explicit group is separable, stably finite, and non-MF.

1 INTRODUCTION

Let $(d_n)_{n \in \mathbb{N}}$ be a sequence of positive integers and let

$$(1) \quad \mathcal{Q} = \prod_n M_{d_n}(\mathbb{C}) / \bigoplus_n M_{d_n}(\mathbb{C})$$

be the associated *norm matrix corona*: bounded sequences of matrices modulo sequences converging to 0 in operator norm. Following Blackadar–Kirchberg [BK], a separable C^* -algebra is *MF* if it embeds into some algebra of the form (1). We use *MF group* exclusively in the sense introduced by Carrión–Dadarlat–Eckhardt and used by Shulman [CDE, Sh]: a countable discrete group E is MF if, for some strictly increasing sequence of positive integers (d_n) , there is an injective homomorphism

$$E \longrightarrow \mathcal{U}(\mathcal{Q}).$$

We shall use the equivalent coordinate model

$$(2) \quad \mathcal{U}_{\mathrm{cor}}((d_n)) := \prod_n \mathcal{U}(M_{d_n}(\mathbb{C})) / \{(u_n) : \|u_n - 1\| \longrightarrow 0\}.$$

The denominator is normal, and polar correction gives a canonical isomorphism

$$(3) \quad \kappa_{(d_n)} : \mathcal{U}_{\mathrm{cor}}((d_n)) \xrightarrow{\cong} \mathcal{U}(\mathcal{Q}),$$

proved in Lemma 6.2. Thus one may equivalently embed E into $\mathcal{U}_{\mathrm{cor}}((d_n))$.

Proposition 1.1 (equivalent MF formulations). *For every countable group, the genuine C^* -corona definition above is equivalent to the unitary-sequence definition. The latter is equivalent to operator-norm local models with separation constant 1, and allowing arbitrary positive dimensions is equivalent to requiring the dimensions to be strictly increasing.* [Lean: GroupApproximation.M
manuscriptExactWrappers.manuscriptMFDefinitionEquivalences]

Proof. Lemma 6.2 identifies the unitary-sequence quotient with the unitary group of the actual C^* -quotient. Replacing an arbitrary finite coordinate set by a standard set of the same cardinality merely conjugates that coordinate by a permutation unitary. This proves the first equivalence once the dimension convention is normalized.

Suppose first that $\rho: G \rightarrow \mathcal{U}_{\text{cor}}((d_n))$ is injective, and choose unitary lifts $u_n(g)$ of $\rho(g)$. Fix a finite set $F \subseteq G$ and $\varepsilon > 0$. For each ordered pair $g \neq h$ in F , injectivity says that $(u_n(g)^{-1}u_n(h))_n$ does not converge to the identity in operator norm. Hence there are $\delta_{g,h} > 0$ and arbitrarily large coordinates at which

$$\|u_n(g) - u_n(h)\| \geq \delta_{g,h}.$$

For each number $\delta_{g,h}$, compactness of

$$\{\lambda \in \mathbb{T} : |1 - \lambda| \geq \delta_{g,h}\}$$

gives an integer $M_{g,h}$ such that every point of this set has a power $1 \leq m \leq M_{g,h}$ whose distance from 1 is greater than 1. The quotient homomorphism law makes all multiplicative defects for pairs in F arbitrarily small at every sufficiently large coordinate. Choose, for each $g \neq h$, a coordinate with the displayed separation and with every required defect at most $\varepsilon/M_{g,h}$; then choose the corresponding power $m \leq M_{g,h}$. The m th tensor power separates the designated pair by more than 1, while the tensor-product telescoping identity bounds each new defect by $m\varepsilon/M_{g,h} \leq \varepsilon$. Finally take the block sum of these finitely many pair-specific models. Operator norms of block diagonal matrices are maxima, so every pair in F is separated by at least 1 and every required product has defect at most ε . This gives the normalized local model.

Conversely, enumerate the countable group and choose increasing finite sets F_n exhausting it. Apply the local condition to F_n with errors tending to zero. For each fixed g, h , the resulting unitary maps are eventually multiplicative on g, h , so their classes define a homomorphism into the unitary-sequence quotient. If $g \neq h$, both eventually lie in F_n and their distance is at least 1; their quotient classes are therefore distinct. The homomorphism is injective.

It remains only to normalize dimensions. Given positive dimensions d_n , choose recursively

$$D_0 = d_0 + 1, \quad D_{n+1} \geq \max\{d_{n+1} + 1, D_n + 1\}.$$

Pad every coordinate model by an identity block:

$$u_n(g) \mapsto u_n(g) \oplus 1_{D_n - d_n}.$$

The dimensions D_n increase strictly. Products, multiplicative defects, and pairwise operator-norm distances are unchanged, because the added block is the same identity representation for every group element. Thus the construction preserves and reflects the quotient relation and injectivity. Forgetting strict increase proves the converse. $\square$

Hereafter “MF” means this equivalent actual-corona/local-operator-norm property. It should not be confused with the stronger reduced-norm convention called MF in [Sc, GKEMP], where approximately multiplicative maps must also recover the canonical trace and the left-regular norm on every group-ring element, or with purely MF (PMF), where those strongly convergent models are genuine finite-dimensional representations [MdlS, Definition 1.2]. Every assertion below uses only the MF property fixed here. The conceptual result is the following explicit finitely presented Kazhdan–Clifford construction.

The rigidity mechanism is best remembered before the construction. The same elementary principle appears in three finite worlds:

$$\begin{array}{ccc} X \subseteq Y, |X| = |Y| & V \subseteq W, \dim V = \dim W & \begin{array}{c} p \leq q, p \sim q \\ \text{in a finite } C^*- \text{algebra} \end{array} \\ \Downarrow & \Downarrow & \Downarrow \\ X = Y & V = W & p = q. \end{array}$$

Finite-dimensional representations use the middle implication directly. For norm-matrix coronas we apply that same implication to spectral projections in each matrix coordinate, with estimates uniform in the coordinate size. This yields the reusable theorem behind every result below.

Theorem 1.2 (Kazhdan transport of asymptotic commutants). *Let Γ and H be countable groups, let Γ have property (T), let $\iota: \Gamma \rightarrow H$ be a homomorphism, let $s \in H$, and let (d_n) be a sequence of positive integers. Suppose*

$$s\iota(\Gamma)s^{-1} \subseteq \iota(\Gamma).$$

Let $U_n: H \rightarrow \mathcal{U}(d_n)$ be an operator-norm asymptotic representation. If (x_n) is uniformly operator-norm bounded and

$$\|[x_n, U_n(\iota(\gamma))]\|_2 \longrightarrow 0 \quad (\gamma \in \Gamma),$$

then

$$\|[U_n(s)x_nU_n(s)^*, U_n(\iota(\gamma))]\|_2 \longrightarrow 0 \quad (\gamma \in \Gamma).$$

[Lean: GroupApproximation.KazhdanAsymptoticCommutant.manuscriptKazhdanTransport]

Proof. Argue by contradiction: fix $\gamma_0 \in \Gamma$ and pass to a subsequence on which the asserted commutator norms for γ_0 stay bounded away from zero. Fix a free ultrafilter ω on that subsequence, put $K_n = L^2(M_{d_n}(\mathbb{C}), \text{tr}_{d_n})$, and form the Hilbert ultraproduct K_ω . The adjoint actions define an honest unitary representation

$$\pi(g) = (\text{Ad } U_n(g))_\omega \quad (g \in H)$$

on K_ω . They also belong to the norm ultraproduct

$$B_\omega = \prod_{\omega} B(K_n).$$

Its natural action on K_ω is faithful (choose coordinate unit vectors which almost attain the operator norms). Moreover B_ω is finite: if $v^*v = 1$, polar correction replaces any representing sequence for v by unitaries along ω , so $vv^* = 1$.

Let $p_\Gamma \in C_{\max}^*(\Gamma)$ be the Kazhdan projection [BHV], let $\pi_\Gamma = \pi \circ \iota$, put $P = \pi_\Gamma(p_\Gamma) \in B_\omega$ using the integrated form of π_Γ , and write $V = \pi(s)$. The range of P is the fixed space $K_\omega^{\iota(\Gamma)}$. The one-sided compression gives

$$PK_\omega = K_\omega^{\iota(\Gamma)} \subseteq K_\omega^{s\iota(\Gamma)s^{-1}} = VPK_\omega, \quad \text{so} \quad P \leq VPV^*.$$

The two projections are unitarily equivalent in the finite algebra B_ω ; hence

$$\boxed{P = VPV^*}.$$

Uniform boundedness makes $x = (x_n)_\omega$ a vector in K_ω , and the hypothesis says $x \in PK_\omega$. Therefore $Vx \in PK_\omega$, contrary to the choice of γ_0 and the subsequence. This proves the ordinary, rather than merely ultrafilter, convergence asserted in the theorem. $\square$

The finite-dimensional reason is equally short. A Kazhdan spectral projection P_n is carried by the compressor to a unitarily equivalent projection Q_n . One-sided compression gives $\|(1 - Q_n)P_n\| \rightarrow 0$, while equality of the two finite ranks reverses the leakage:

$$(4) \quad \boxed{\text{rank } P_n = \text{rank } Q_n, \quad \|(1 - Q_n)P_n\| \rightarrow 0 \implies \|(1 - P_n)Q_n\| \rightarrow 0.}$$

The two estimates transport the asymptotic fixed space. The uniform spectral-cut argument is given in Section 7.

Central-corner reduction. Let $\Theta: H \rightarrow \mathcal{U}(\mathcal{Q})$ be a corona representation and let $0 \neq q \in \mathcal{Q}$ be a projection commuting with $\Theta(H)$. Then $\Theta_q(g) = q\Theta(g)q$ is a homomorphism into the unitary group of the corner $q\mathcal{Q}q$, whose unit is q , and this corner is itself a positive-rank norm matrix corona. Indeed, a spectral cut at $1/2$ of a self-adjoint lift of q gives coordinate projections (q_n) representing q . Put $r_n = \text{rank } q_n$. Nonzeroness makes $J = \{n : r_n > 0\}$ infinite. Compression identifies $q_n M_{d_n}(\mathbb{C}) q_n$ with $M_{r_n}(\mathbb{C})$ on J ; the zero coordinates contribute nothing, and relabeling J by $\mathbb{N}$ identifies $q\mathcal{Q}q$ with a corona of positive matrix ranks. Compression of Θ is multiplicative because q is central relative to its image, and unitary coordinate lifts follow by polar correction.

The entire non-MF argument is the following two-element consequence.

Theorem 1.3 (central-sign criterion). *Let H and Γ be countable groups, let Γ have property (T), let $\iota: \Gamma \rightarrow H$ be a homomorphism, and suppose*

$$t\iota(\Gamma)t^{-1} \subseteq \iota(\Gamma), \quad [c, \iota(\Gamma)] = 1.$$

Put $d = tct^{-1}$ and $u = [d, \iota(a)]$ for some $a \in \Gamma$. If

$$z = u^2 \neq 1, \quad z^2 = 1, \quad z \in Z(H),$$

then every homomorphism from H to a norm matrix corona kills z , and H is not MF. [Lean: GroupApproximation.KazhdanCompressionCore.manuscriptCentralSignatureCriterion]

Proof. Suppose a corona representation Θ detects z . The nonzero projection, central relative to $\Theta(H)$,

$$q = \frac{1 - \Theta(z)}{2}$$

cuts it to a representation in which $\Theta(z) = -1$. The central-corner reduction above regards this as a representation into a positive-rank norm matrix corona. The element c belongs to the asymptotic commutant of $\iota(\Gamma)$, so Theorem 1.2 puts $d = tct^{-1}$ there as well. Hence $u = [d, \iota(a)] \rightarrow_2 1$, and therefore $z = u^2 \rightarrow_2 1$. On the q -corner, however, $z = -1$, whose normalized Hilbert–Schmidt distance from 1 is 2. This contradiction kills z in every corona representation; since $z \neq 1$, no such representation is faithful. $\square$

This is the proof to keep in mind.

Given Γ, α, a as in the theorem below, the construction is most readable with one redundant generator ε : declare it to be a central involution and impose

$$(5) \quad \boxed{\varepsilon = u^2, \quad u = [tct^{-1}, a].}$$

This is the conceptual presentation: rigidity forces $u \rightarrow_2 1$, whereas the Clifford model makes $u^2 = -1$. The theorem below uses the Tietze-eliminated version. Since $d = tct^{-1}$ is an involution, $u^2 = [d, ada^{-1}]$; conversely, centrality of u^2 forces $(u^2)^2 = 1$. Thus the redundant sign costs one generator and saves the reader the only opaque word in the construction.

Theorem 1.4 (Kazhdan–Clifford construction). *Let Γ be a finitely presented group with property (T), let $\alpha: \Gamma \hookrightarrow \Gamma$ be an injective endomorphism, and choose $a \in \Gamma \setminus \alpha(\Gamma)$. Adjoin elements t, c and impose*

$$t\gamma t^{-1} = \alpha(\gamma), \quad c^2 = 1, \quad [c, \gamma] = 1 \quad (\gamma \in \Gamma).$$

Put

$$d = tct^{-1}, \quad u = [d, a], \quad w = u^2 = [d, ada^{-1}],$$

and impose centrality of w . More explicitly, for a finite generating set S_Γ it is enough to impose the compression and lamp relations for $s \in S_\Gamma$, together with

$$[w, s] = 1 \quad (s \in S_\Gamma), \quad [w, t] = [w, c] = 1.$$

Thus w commutes with a generating set of the entire resulting group. The group $E(\Gamma, \alpha, a)$ is finitely presented, the canonical map $\Gamma \rightarrow E(\Gamma, \alpha, a)$ is injective, and w is a nontrivial central involution. Every homomorphism

$$E(\Gamma, \alpha, a) \longrightarrow \mathcal{U}(\mathcal{Q})$$

to the unitary group of every norm matrix corona kills w . Consequently $E(\Gamma, \alpha, a)$ is not MF. [Lean: GroupApproximation.KazhdanCliffordConstruction.kazhdanCliffordConstruction]

Proof. Finite presentability follows from a relative finite presentation: start with the free product of Γ and the free group on t, c , and quotient by the finitely many displayed relations for a finite generating set of Γ .

The relation $w^2 = 1$ is redundant. Put $b = ada^{-1}$. Both d and b are involutions, $w = [d, b] = (db)^2$, and direct cancellation gives

$$dwd^{-1} = w^{-1}.$$

Centrality also gives $dwd^{-1} = w$, whence $w = w^{-1}$ and $w^2 = 1$.

For nontriviality, form the ascending HNN extension

$$G = \langle \Gamma, t \mid t\gamma t^{-1} = \alpha(\gamma) \rangle$$

and let $X = G/\Gamma$. The mapping-telescope model of this extension embeds Γ . Let G permute the Clifford generators $(e_x)_{x \in X}$ and map c to e_Γ . Then

$$d \mapsto e_{t\Gamma}, \quad ada^{-1} \mapsto e_{at\Gamma}.$$

The two cosets are distinct: equality would give $t^{-1}at \in \Gamma$, equivalently $a \in \alpha(\Gamma)$. Hence the two Clifford generators anticommute and

$$w \mapsto (e_{t\Gamma}e_{at\Gamma})^2 = -1.$$

The Clifford sign is central, so the map respects every defining relation. It follows simultaneously that $w \neq 1$ and that the canonical map of Γ is injective.

Finally, $w = u^2$ is a nontrivial central involution. The central-sign criterion therefore says that every corona representation kills w . Since $w \neq 1$, none is faithful. $\square$

The sign argument has a useful Clifford-free strengthening. For a subgroup $L \leq H$, let $G_{\text{comp}}(L)$ be generated by the elements s satisfying $sLs^{-1} \subseteq L$, and put

$$(6) \quad \mathfrak{D}(H, L) = \left\langle\left\langle [gzg^{-1}, \ell] : g \in G_{\text{comp}}(L), z \in C_H(L), \ell \in L \right\rangle\right\rangle_H.$$

Theorem 1.5 (compression radical). *Let H and Γ be countable groups, let Γ have property (T), let $\iota: \Gamma \rightarrow H$ be a homomorphism, and put $L = \iota(\Gamma)$. If $F \trianglelefteq H$ is finite and $F \leq \mathfrak{D}(H, L)$, then every homomorphism from H to every norm matrix corona kills F .* [Lean: GroupApproximation.KazhdanAsymptoticCommutant.manuscriptCompressionRadical]

Indeed, the equality $P = VPV^*$ makes transport invariant under both a compressor and its inverse; iteration therefore handles every $g \in G_{\text{comp}}(L)$. Transport makes every displayed generator of $\mathfrak{D}(H, L)$ Hilbert–Schmidt trivial, hence does the same to its normal closure. If a corona representation

detected F , the complementary Reynolds corner would make $\sum_{f \in F} U_f$ converge to zero in operator norm, while transport would make it converge to $|F|1$ in normalized Hilbert–Schmidt norm. This is impossible.

The same element a drives both halves of the proof. Analytically, $a \in \Gamma$, so fixed-space transport forces the displaced lamp to commute with its image in finite-dimensional norm models. Geometrically, $a \notin \alpha(\Gamma)$, so a moves that lamp to a different HNN coset, where Clifford anticommutation produces -1 . Thus a lives on the boundary of the self-compression: rigidity cannot detect that boundary, while the Clifford representation detects it by a sign.

The following theorem is the fully explicit finite presentation developed in the second half of the paper.

Theorem A (explicit Kazhdan–Clifford group). *Let E be the group on the eight generators $v_1, v_2, v_3, x, y, z, t, c$ with the finite list of relators in Definition 3.1. Put*

$$w = [tct^{-1}, v_1(tct^{-1})v_1^{-1}].$$

The defining relations make w central; together with $c^2 = 1$, centrality implies $w^2 = 1$. Then:

- (1) $w \neq 1$ in E , so w is a central involution;
- (2) for every sequence (d_n) of positive integers and every group homomorphism $\Theta: E \rightarrow \mathcal{U}_{\text{cor}}((d_n))$ into the corresponding unitary norm corona (2), one has $\Theta(w) = 1$.

In particular E is not MF, and neither $C_{\text{max}}^(E)$ nor $C_{\text{red}}^*(E)$ is an MF C^* -algebra. Here $C_{\text{max}}^*(E)$ denotes the universe-relative bounded-product model specified in Proposition 2.1.* [Lean: GroupApproximation.LiteralNonMFEndpoint.manuscriptTheoremA]

A completely explicit presentation of E — eight generators, with every relation displayed — appears as (9) in Definition 3.1.

For the proof, put $d = tct^{-1}$ and $u = [d, v_1]$. Lemma 8.1 gives the useful identity $w = u^2$. The proof has two opposed halves, both centered on u :

$$\boxed{\text{matrix-corona rigidity: } u \sim_2 1} \quad \text{and} \quad \boxed{\text{Clifford model: } u^2 = -1.}$$

For a corona representation that detects w , cut to the negative central corner of w . Kazhdan transport gives $u \sim_2 1$, hence $w = u^2 \sim_2 1$; the corner identity gives $w = -1$. This contradiction is the analytic core of Theorem A. Section 7 proves the finite-normal criterion and its coordinate estimates, and Section 8 applies them to the literal group. [Lean: GroupApproximation.LiteralNonMFEndpoint.negativeCorner_kazhdanTransport_contradiction]

Nontriviality of w is witnessed by a completely explicit representation of E into a semidirect product of a Clifford group by a mapping-telescope group built from exact affine matrices over $\mathbb{Z}[1/2]$ (Section 4); no combinatorial theory of HNN extensions is needed.

Two companion results clarify the mechanism. First, in finite dimensions the obstruction is elementary and needs neither property (T), nor unitarity, nor centrality of w :

We use the intrinsic compression–centralizer defect $\mathfrak{D}(H, L)$ defined above.

Theorem 1.6 (finite-dimensional shadow). *Let k be a field, V a finite-dimensional k -vector space, H a group, $\Gamma \leq H$ a subgroup, and $t, c \in H$, $a \in \Gamma$ elements with $t\Gamma t^{-1} \subseteq \Gamma$ and $c\gamma = \gamma c$ for all $\gamma \in \Gamma$. Then every homomorphism $\pi: H \rightarrow \mathrm{GL}(V)$ satisfies*

$$\pi([tct^{-1}, a(tct^{-1})a^{-1}]) = 1.$$

More generally, every such π sends the entire subgroup $\mathfrak{D}(H, \Gamma)$ to the identity.

[Lean: GroupApproximation.ManuscriptExactWrappers.manuscriptTheoremB]

Applied to $H = E$, Theorem 1.6 shows that *every* finite-dimensional representation of E over *every* field sends w to the identity: finite-dimensional linear representations do not separate the points of E , no finite quotient can witness $w \neq 1$, and the infinite Clifford witness of Section 4 cannot be replaced by a finite-dimensional one. Property (T) enters the corona argument through a dimension-free spectral gap for the Hermitian adjoint averages. It supplies one cutoff and uniform estimates for the finite-coordinate projections P_n , independently of the matrix dimensions.

Second, the elementary finite-dimensional calculation does not determine whether its marked word is trivial:

Proposition 1.7 (cyclic-base comparison). *Define the literal marked presentation*

$$E_{\mathrm{BS}} = \left\langle \gamma_0, t, c \left| \begin{array}{l} t\gamma_0 t^{-1} = \gamma_0^2, \quad c^2 = 1, \quad [c, \gamma_0] = 1, \\ w_{\mathrm{BS}}^2 = 1, \quad [w_{\mathrm{BS}}, \gamma_0] = [w_{\mathrm{BS}}, t] = [w_{\mathrm{BS}}, c] = 1 \end{array} \right. \right\rangle,$$

where $w_{\mathrm{BS}} = [tct^{-1}, \gamma_0(tct^{-1})\gamma_0^{-1}]$. Then E_{BS} is finitely presented and $w_{\mathrm{BS}} \neq 1$. There is an explicit surjective homomorphism from E_{BS} onto its realized Clifford quotient that maps w_{BS} to the nontrivial central sign. Nevertheless every finite-dimensional representation of E_{BS} over every field maps w_{BS} to the identity.

[Lean: GroupApproximation.LiteralCyclicCalibration.manuscriptCyclicCalibration]

The quotient statement makes the comparison at the level actually used by the proof: exact finite-dimensional invisibility does not by itself force triviality of the marked word. The MF conclusion in Theorem A is the separate uniform assertion supplied by the fixed-space argument.

Generalizations and consequences Theorem A is an instance of a general finite-normal obstruction (Definition 7.1). A marked Kazhdan pattern consists of a property (T) group Λ , a homomorphism $\iota: \Lambda \rightarrow H$, and elements $t, c \in H$ such that $t\iota(\Lambda)t^{-1} \subseteq \iota(\Lambda)$ and $[c, \iota(\Lambda)] = 1$. Theorem 7.2 sends every finite normal subgroup contained in the resulting compression-defect subgroup to the identity under every corona representation. For E , the relevant subgroup is $\{1, w\}$. Corollary 9.5 records the resulting obstruction for C^* -subalgebras of norm matrix coronas, while Theorem 1.7 shows that finite-dimensional collapse by itself does not decide whether the marked word is trivial.

Section 9 develops structural and C^* -algebraic consequences. Its principal C^* -algebraic application is the following explicit reduced group algebra:

Theorem B (the reduced group algebra of E). *$C_{\text{red}}^*(E)$ is a unital separable C^* -algebra which carries a faithful tracial state — and is therefore stably finite — but is not an MF algebra.* [Lean: GroupApproximation.LiteralNonMFEndpo
int.manuscriptTheoremD]

Operator norm versus tracial approximation The finite projection argument is specific to operator norm. In a tracial ultraproduct, normalized Hilbert–Schmidt-small leakage does not retain the coordinate ranks needed for the equal-rank reversal, so the argument below does not by itself give a hyperlinear obstruction.

External inputs The numbered core uses two group-theoretic inputs: the displayed presentation of $\text{SL}_3(\mathbb{Z})$ and the classical fact that $\mathbb{Z}^3 \rtimes \text{SL}_3(\mathbb{Z})$ has property (T). The exact twenty-relator base is identified with that affine group in Section 3. The Clifford witness and the finite-dimensional obstruction are proved directly.

Relation to prior work The closest conceptual analogue is the recent work on nonsofic groups. One-sided compression of a property (T) subgroup and propagation from its centralizer appear in [OAI, KT], building on rigidity results of Kun and Kun–Thom [Ku16, KT19]; related centralizer rigidity for sofic embeddings is proved in [AT]. Those results concern Hamming or tracial approximations. Here coordinatewise averaging, polar correction, and a finite-stage spectral projection produce a specific element in the kernel of every operator-norm corona representation. The Clifford central extension supplies the element whose survival is proved independently.

The operator-norm argument replaces finite cardinality by the implication

$$\begin{aligned} \text{one-sided leakage of } P_n \text{ into } R_n, \quad \text{rank } P_n = \text{rank } R_n \\ \implies \|P_n - R_n\| \longrightarrow 0. \end{aligned}$$

The complementary finite-group corner then converts this estimate into a fixed nontrivial element sent to the identity by every corona representation.

Slofstra’s constructions supply close precedents for the Clifford group, distinguished central involution, shift, and HNN doubling map [Slo18, Section 2 and Remark 3.3]. Negative-eigenspace compression appears in Slofstra–Vidick [SV, proof of Proposition 2.7], and Bachner–Dogon–Lubotzky develop the operator-norm rounding, compression, and polar-correction procedure in detail [BDL, Lemmas 2.2–2.3, Proposition 2.4, and the proof of Proposition 1.5]. The present proof combines that local corner procedure with the one-sided Kazhdan compressor; equality of the two finite ranks supplies the reverse projection estimate and hence the universal kernel conclusion.

The analytic passage from operator norm to the Hilbert–Schmidt conjugation representation belongs to the $\pi \otimes \bar{\pi}$ framework of Bekka and Bekka–Valette [Be, BV]. Dadarlat uses normalized finite-rank projections as almost

invariant Hilbert–Schmidt vectors in an operator-norm approximation argument [Dad, Lemma 3.18 and Proposition 3.19]. These are the direct analytic precedents for the conjugation step used here.

Slofstra’s quotients by elements invisible to exact or approximate finite-dimensional representations precede the MF-radical language used below [Slo17, Definitions 2.5–2.7]. Fritz’s marked $BS(2, 3)$ example provides additional context for the cyclic comparison [Fri, SV].

Central extensions and rigidity of approximate representations also drive the Frobenius obstruction of De Chiffre–Glebsky–Lubotzky–Thom [DGLT]. In the proof of Proposition 1.6, Bachner–Dogon–Lubotzky treat the finite-central case by a character-isotypic corner, and reduce the general finite-normal case to it by passing to the finite-index centralizer [BDL]. The finite-normal criterion here combines the corresponding coordinate-corner construction with the one-sided Kazhdan compressor and does not assume that the quotient is hyperlinear. Related finite-target approximation and extension results for weak soficity appear in [GR, GI].

Carrión–Dadarlat–Eckhardt provide the group-MF framework used here and, through Abels’ group, an amenable MF group that is not residually finite [CDE, Section 2.14 and Corollary 2.18]. The cyclic comparison addresses the narrower question of what happens to the marked compression pattern when property (T) is removed.

Finally, failure of the MF property for a group C^* -algebra does not by itself prevent the group from embedding into a corona: the group property requires only a faithful canonical image in some MF quotient of $C_{\max}^*(E)$. The result here produces a specific nontrivial element in the kernel of every such representation. For broader context see [DGLT, BDL, OAI, KT, Sh, Dad, Slo17, Slo18, SV].

Organization Section 2 fixes notation and states the external inputs. Section 3 constructs E and proves property (T) for its literal base. Section 4 proves $w \neq 1$. Section 5 proves Theorem 1.6. Section 6 develops the corona lemmas. Section 7 defines marked Kazhdan patterns and their compression-defect subgroups, proves the finite-normal obstruction criterion (Theorem 7.2), and develops the coordinate Reynolds and Kazhdan-compression estimates. Section 8 proves the defect-square identity and deduces Theorem A. Section 9 derives Theorem B and further consequences. Section 10 records the cyclic comparison as a secondary exact-dimensional shadow. Section 11 explains the operator-norm boundary of the argument.

2 NOTATION AND EXTERNAL INPUTS

All groups are discrete and countable. Property (T) is the textbook complex-unitary property. A C^* -homomorphism is unital only when this is stated. In particular, an MF embedding may be nonunital, as is customary for embedding a unital algebra as a C^* -subalgebra. If $e: A \rightarrow B$ is an injective, possibly nonunital $*$ -homomorphism between unital C^* -algebras,

then $u \mapsto e(u) + (1 - e(1))$ is an injective homomorphism $\mathcal{U}(A) \rightarrow \mathcal{U}(B)$; this complement correction is used below. For a unital C^* -algebra A we write $\mathcal{U}(A)$ for its unitary group. On $M_r(\mathbb{C})$ we use the normalized trace tr_r , the operator norm $\|\cdot\|$, and the normalized Hilbert–Schmidt norm $\|x\|_2 = \text{tr}_r(x^*x)^{1/2}$. We will use three elementary facts without further comment: $\|x\|_2 \leq \|x\|$; $|\text{tr}_r(x)| \leq \|x\|$; and $\|uxv\|_2 = \|x\|_2$ for unitaries u, v . For a nonempty finite set Y , write $M_Y(\mathbb{C}) = \mathbb{C}^{Y \times Y}$ for the corresponding indexed matrix algebra.

Given a sequence $(d_n)_{n \in \mathbb{N}}$ of positive integers, we write

$$\mathcal{Q}((d_n)_{n \in \mathbb{N}}) = \prod_{n \in \mathbb{N}}^{\ell^\infty} M_{d_n}(\mathbb{C}) / \bigoplus_{n \in \mathbb{N}}^{c_0} M_{d_n}(\mathbb{C}),$$

the quotient of the bounded C^* -product by the ideal of sequences with $\|x_n\| \rightarrow 0$. We call any such algebra a *norm matrix corona*. We also write

$$\mathcal{U}_{\text{cor}}((d_n)_{n \in \mathbb{N}}) = \prod_{n \in \mathbb{N}} \mathcal{U}(M_{d_n}(\mathbb{C})) / \{(u_n) : \|u_n - 1\| \rightarrow 0\}.$$

A *corona representation* of a group E is a homomorphism $E \rightarrow \mathcal{U}(\mathcal{Q})$. Lemma 6.2 canonically identifies it with a homomorphism into $\mathcal{U}_{\text{cor}}((d_n))$, which we call its coordinate model. The group E is *MF* if it admits an injective corona representation for a strictly increasing dimension sequence. By the dimension-amplification equivalence discussed in the introduction, allowing an arbitrary positive dimension sequence gives the same class of groups.

Fix a set-theoretic universe $\mathfrak{U}$. For the following construction, let H and all representation targets be $\mathfrak{U}$ -small; countability is not needed. Define $C_{\text{max}, \mathfrak{U}}^*(H)$ as the closed $*$ -subalgebra generated by the diagonal copy of H in the bounded product of all its nontrivial unitary representations on $\mathfrak{U}$ -small unital C^* -algebras. This algebra lives in the next universe. We suppress the subscript $\mathfrak{U}$ below and write u_h for its canonical unitary associated with $h \in H$.

Proposition 2.1 (universe-relative maximal group algebra). *The canonical map $h \mapsto u_h$ is injective. For every $\mathfrak{U}$ -small unital C^* -algebra B and every homomorphism $\rho: H \rightarrow \mathcal{U}(B)$, there is a unique unital $*$ -homomorphism $C_{\text{max}}^*(H) \rightarrow B$ that maps u_h to $\rho(h)$ for every $h \in H$.* [Lean: GroupApproximation.manuscriptUniverseRelativeMaximalGroupCStar]

For a nontrivial target, projection to its coordinate gives the extension; for a trivial target, the map is unique. Density of the generated $*$ -subalgebra gives uniqueness, while the left-regular coordinate proves injectivity.

We use the commutator convention $[x, y] = xyx^{-1}y^{-1}$.

3 THE LITERAL GROUP AND ITS KAZHDAN SOURCE

We work throughout with the groups defined by the relators below. The proof of property (T) identifies the base with a standard affine matrix group;

every later relation involving its image in E is used directly from the displayed presentation. First define the *rotation group*

$$(7) \quad \mathcal{R} = \left\langle x, y, z \mid \begin{array}{l} x^3 = y^3 = z^2 = (xz)^3 = (yz)^3 = 1, \\ (x^{-1}zxy)^2 = (y^{-1}zyx)^2 = (xy)^6 = 1 \end{array} \right\rangle,$$

and the *literal base*

$$(8) \quad \mathcal{B} = \left\langle v_1, v_2, v_3, x, y, z \mid \begin{array}{l} \text{the eight relations in (7),} \\ [v_1, v_2] = [v_1, v_3] = [v_2, v_3] = 1, \\ xv_1x^{-1} = v_3, \quad xv_2x^{-1} = v_1, \quad xv_3x^{-1} = v_2, \\ yv_1y^{-1} = v_1, \quad yv_2y^{-1} = v_2^{-1}v_3, \quad yv_3y^{-1} = v_1v_2^{-1}, \\ zv_1z^{-1} = v_2v_3^{-1}, \quad zv_2z^{-1} = v_1v_3^{-1}, \quad zv_3z^{-1} = v_3^{-1} \end{array} \right\rangle.$$

Write $\iota: \mathcal{B} \rightarrow E$ for the homomorphism induced by the first six generators of the presentation below. This map is not assumed to be injective.

Definition 3.1. Use the six literal base generators and the relations displayed in (8). Define

$$(9) \quad E = \left\langle v_1, v_2, v_3, x, y, z, t, c \mid \begin{array}{l} \text{all relations in (8),} \\ tv_it^{-1} = v_i^2 \quad (1 \leq i \leq 3), \\ txt^{-1} = x, \quad tyt^{-1} = y, \quad zt^{-1} = z, \\ c^2 = 1, \\ [c, g] = 1 \quad (g \in \{v_1, v_2, v_3, x, y, z\}), \\ [w, g] = 1 \quad (g \in \{v_1, v_2, v_3, x, y, z, t, c\}) \end{array} \right\rangle,$$

where w abbreviates the word $[tct^{-1}, v_1(tct^{-1})v_1^{-1}]$. This is an explicit finite presentation with eight generators. Its defining relations make w central, and centrality together with $c^2 = 1$ implies $w^2 = 1$ by the calculation in the proof of Theorem 1.4; the relations on the six base generators also give

$$t\iota(\mathcal{B})t^{-1} \subseteq \iota(\mathcal{B}), \quad [c, \iota(\mathcal{B})] = 1.$$

[Lean: GroupApproximation.LiteralNonMFPresentation.manuscriptLiteralPresentation]

Proposition 4.3 below proves that $w \neq 1$. Figure 1 isolates the geometry used later in the analytic argument: $d = tct^{-1}$ centralizes the compressed copy, not necessarily all of $\iota(\mathcal{B})$, and the marked commutator measures precisely that failure after conjugating d by $\iota(v_1)$.

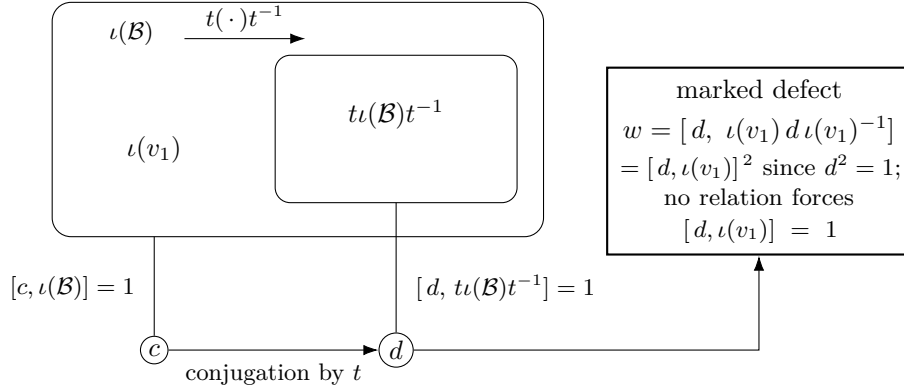


Figure 1: The marked-compression pattern. Conjugation by t carries the image of the literal base into itself. It sends c to $d = tct^{-1}$: the relations force c to commute with all of $\iota(\mathcal{B})$ and force d to commute with the compressed copy, but do not require d to commute with $\iota(v_1)$. The marked word w records that commutation defect. Only the one-sided inclusion is used by the analytic criterion; the separate Clifford witness proves that the canonical base map in this explicit group is injective.

In particular the compression argument needs only the one-sided inclusion and centralizer relations shown in Figure 1; it never uses injectivity or properness of the base map. Injectivity for the constructed group is established independently by the Clifford realization.

3.1 Property (T) of the base

Proposition 3.2. *The twenty-relator group $\mathcal{B}$ has property (T), in both the real-orthogonal and complex-unitary formulations.* [Lean: GroupApproximation.L
literalBaseP13PropertyTBridge.manuscriptBaseHasKazhdanPropertyT]

Proof. The eight relations in (7) are the Conder–Robertson–Williams presentation of $\mathrm{SL}_3(\mathbb{Z})$ [CRW, Theorem 2]. The matrix-labelled form for x, y, z displayed in Lemma 4.2 is also recorded in [CLV]. The remaining relations say that v_1, v_2, v_3 commute and that x, y, z act on them by the columns of those three matrices. For completeness, this does more than define a quotient. The map from the presentation to $\mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z})$ sends v_i to $(e_i, 1)$ and each rotation generator to its labelled matrix. In the other direction, send (n_1, n_2, n_3, A) to

$$v_1^{n_1} v_2^{n_2} v_3^{n_3} \overline{A},$$

where $A \mapsto \overline{A}$ is the inverse of the isomorphism from the eight-relator rotation group to $\mathrm{SL}_3(\mathbb{Z})$. The nine conjugation relations give covariance first for x, y, z , hence for every A ; so this second map is a homomorphism. The two maps fix the six displayed generators and are inverse.

Property (T) of $\mathbb{Z}^n \rtimes \mathrm{SL}_n(\mathbb{Z})$ for $n \geq 3$ is classical [BHV, Example 1.7.4(i)]. Realification and complexification identify the real and complex formulations. $\square$

4 THE WITNESS: $w \neq 1$

Clifford signs, shifts, and HNN doubling have appeared together before in approximate-representation theory, notably in Slofstra’s hyperlinear-profile construction [Slo18, Section 2]. The construction below uses the Clifford lamp only as an explicit algebraic witness. It uses the displayed quotient of the literal base and is independent of the semantic identification used above to prove property (T).

Construction 4.1 (Clifford lamp groups). For a set X , use the real Clifford-algebra convention generated by $(c_x)_{x \in X}$ with $c_x^2 = 1$ and $c_x c_y = -c_y c_x$ for $x \neq y$, and let $\mathrm{CLamp}(X)$ be the subgroup of units generated by -1 and the c_x . Write $\zeta = -1$. In this concrete group,

$$\zeta^2 = c_x^2 = 1, \quad c_x c_y = \zeta c_y c_x \quad (x \neq y).$$

The sign ζ is nontrivial. Every permutation of X acts on this group by $c_x \mapsto c_{\sigma(x)}$ and fixes ζ . [Lean: GroupApproximation.ManuscriptExactWrappers.manuscriptCliffordConstruction]

Lemma 4.2 (linear model). *For $M \in \mathrm{GL}_3(\mathbb{Q})$ and $u \in \mathbb{Q}^3$, write*

$$A(M, u) = \begin{pmatrix} M & u \\ 0 & 1 \end{pmatrix}.$$

Assign $v_i \mapsto A(I_3, e_i)$ and

$$\begin{aligned} x &\mapsto A\left(\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}, 0\right), & y &\mapsto A\left(\begin{pmatrix} 1 & 0 & 1 \\ 0 & -1 & -1 \\ 0 & 1 & 0 \end{pmatrix}, 0\right), \\ z &\mapsto A\left(\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ -1 & -1 & -1 \end{pmatrix}, 0\right). \end{aligned}$$

Let $\bar{\Gamma} \leq \mathrm{GL}_4(\mathbb{Q})$ be the subgroup generated by these six matrices, and put $D = \mathrm{diag}(2, 2, 2, 1)$. Direct matrix computation proves that the six matrices satisfy all twenty relators of $\mathcal{B}$ and that

$$\bar{\alpha}(u) = DuD^{-1}$$

is an injective endomorphism of $\bar{\Gamma}$ which squares the three translation generators and fixes x, y, z . Moreover the matrix assigned to v_1 does not lie in $\mathrm{range}(\bar{\alpha})$. [Lean: GroupApproximation.ManuscriptExactWrappers.manuscriptLinearModel]

Proof. The twenty relator checks reduce to sixteen rational identities for each matrix equation. Conjugation by D gives the six stable-letter equations and is injective because it is conjugation by an invertible matrix. Every displayed generator and its explicitly computed inverse has integral entries;

equivalently, all six generators lie in $\mathrm{GL}_4(\mathbb{Z})$. Hence $\bar{\Gamma} \leq \mathrm{GL}_4(\mathbb{Z})$. However $D^{-1}\bar{v}_1 D$ has entry $1/2$ in position $(1, 4)$, so it is not in $\bar{\Gamma}$. This proves the last assertion. $\square$

Form the mapping telescope of $\bar{\alpha}$, extend it by its shift τ , and call the resulting group V . Let $j: \bar{\Gamma} \hookrightarrow V$ be its level-zero map and put $X = V/j(\bar{\Gamma})$, with root $o = j(\bar{\Gamma})$. The final assertion of Lemma 4.2 is exactly the statement that

$$\tau o \neq j(\bar{v}_1)\tau o.$$

Thus the Clifford generators at these two sites anticommute.

Proposition 4.3. *$w \neq 1$ in E .* [Lean: GroupApproximation.LiteralNonMFLinearWitness.literal_mark_ne_one]

Proof. Form $W := \mathrm{CLamp}(X) \rtimes V$ as in Construction 4.1. Define a map on the generators of (9) by

$$\begin{aligned} g \mapsto j(\bar{g}) \in V \quad (g \in \{v_1, v_2, v_3, x, y, z\}), \\ t \mapsto \tau, \quad c \mapsto c_o. \end{aligned}$$

The exact checks in Lemma 4.2 give the twenty base and six stable-letter relations. The level-zero subgroup fixes o , so c_o commutes with every base generator, and $c_o^2 = 1$. Under this assignment,

$$tct^{-1} \mapsto c_{\tau o}, \quad v_1(tct^{-1})v_1^{-1} \mapsto c_{j(\bar{v}_1)\tau o},$$

and the two sites are distinct by Lemma 4.2, so

$$w \mapsto [c_{\tau o}, c_{j(\bar{v}_1)\tau o}] = -1 = \zeta.$$

Since ζ is central in W , the remaining relations $[w, g] = 1$ for $g \in \{v_1, v_2, v_3, x, y, z, t, c\}$ are also satisfied. Hence the assignment extends to a group homomorphism $E \rightarrow W$ sending w to $\zeta \neq 1$, and therefore $w \neq 1$ in E . $\square$

Scope of the witness. The target uses the infinite group $\mathrm{CLamp}(X)$ when X is infinite, but each group element involves only finitely many Clifford sites; the remaining data are rational matrices. No normal-form theorem for an HNN extension enters: the mapping telescope supplies the required group and the range characterization separates the two sites.

5 THE FINITE-DIMENSIONAL OBSTRUCTION

We prove Theorem 1.6. Everything happens inside the finite-dimensional algebra $\mathrm{End}_k(V)$.

Proof of Theorem 1.6. Let $\pi: H \rightarrow \mathrm{GL}(V)$ be a homomorphism and let

$$C = \{x \in \mathrm{End}_k(V) : \pi(\gamma)x = x\pi(\gamma) \text{ for all } \gamma \in \Gamma\}$$

be the commutant of $\pi(\Gamma)$, a linear subspace of $\mathrm{End}_k(V)$. Let $u = \pi(t)$ and consider the linear automorphism $\Phi(x) = uxu^{-1}$ of $\mathrm{End}_k(V)$.

We claim $C \subseteq \Phi(C)$. Let $x \in C$ and set $y = u^{-1}xu$, so that $x = \Phi(y)$. For $\gamma \in \Gamma$ we have $t\gamma t^{-1} \in \Gamma$, hence $\pi(t\gamma t^{-1})$ commutes with x , and therefore

$$\pi(\gamma)y = u^{-1}\pi(t\gamma t^{-1})xu = u^{-1}x\pi(t\gamma t^{-1})u = y\pi(\gamma).$$

Thus $y \in C$, proving the claim. Since Φ is a linear isomorphism, $\dim_k \Phi(C) = \dim_k C < \infty$, and the inclusion $C \subseteq \Phi(C)$ forces

$$(10) \quad C = \Phi(C) = uCu^{-1}.$$

Now $\pi(c) \in C$ because c centralizes Γ . By (10), $\pi(tct^{-1}) = u\pi(c)u^{-1} \in C$. Since $a \in \Gamma$, the element $\pi(tct^{-1})$ commutes with $\pi(a)$, so $\pi(a(tct^{-1})a^{-1}) = \pi(tct^{-1})$, and therefore

$$\pi([tct^{-1}, a(tct^{-1})a^{-1}]) = [\pi(tct^{-1}), \pi(tct^{-1})] = 1. \quad \square$$

The same calculation proves the final, subgroup-valued assertion of Theorem 1.6. The elements of H whose conjugation action preserves C in both directions form a subgroup. The argument above puts every one-sided compressor in that subgroup, and therefore puts all of $G_{\text{comp}}(\Gamma)$ in it. Thus, for $g \in G_{\text{comp}}(\Gamma)$ and $z \in C_H(\Gamma)$, the operator $\pi(gzg^{-1})$ still belongs to C and commutes with every $\pi(\ell)$, $\ell \in \Gamma$. Hence π kills all the generators in (6), and normality of its kernel kills their normal closure.

Corollary 5.1. *The group E of Definition 3.1 admits no injective homomorphism into $\text{GL}(V)$ for any finite-dimensional vector space V over any field. More strongly, every finite-dimensional linear representation maps w to the identity, and every homomorphism to a finite group maps w to the identity.* [Lean: GroupApproximation.LiteralFiniteDimensionalObstruction.manuscriptFiniteDimensionalConsequences]

Proof. Apply Theorem 1.6 with $H = E$, the image of $\mathcal{B}$, and the elements t, c, v_1 ; the hypotheses hold by the defining relations (9). Every finite-dimensional representation kills w , and $w \neq 1$ by Proposition 4.3. A finite target embeds into a finite symmetric group and hence into a finite-dimensional general linear group via permutation matrices, so it also maps w to the identity. $\square$

Why the corona argument is different. The proof of Theorem 1.6 used only that a finite-dimensional subspace cannot be properly contained in an isomorphic image of itself. An MF model is only asymptotically multiplicative, so its fixed spaces are not literal invariant subspaces. Property (T) repairs this at finite stages: a uniform spectral cut of the adjoint averages captures the almost-fixed vectors. One-sided compression gives a leakage estimate between two equal-rank coordinate projections, and the equal-rank flip supplies the reverse estimate. This is the route used below.

6 NORM MATRIX CORONAS

Throughout this section $\mathcal{Q} = \mathcal{Q}((d_n)_{n \in \mathbb{N}})$ is a norm matrix corona, $q: \prod_{n \in \mathbb{N}}^{\ell^\infty} M_{d_n}(\mathbb{C}) \rightarrow \mathcal{Q}$ the quotient map.

The following lifting facts are standard perturbation theory; see Loring [Lo] for a systematic treatment.

Lemma 6.1 (lifting). *Every unitary $u \in \mathcal{Q}$ lifts to a sequence of unitaries.*

[Lean: GroupApproximation.ManuscriptExactWrappers.manuscriptUnitaryLifting]

Proof. Lift u to an arbitrary bounded sequence (x_n) . Since $q((x_n))$ is unitary, both $\|x_n^* x_n - 1\| \rightarrow 0$ and $\|x_n x_n^* - 1\| \rightarrow 0$. In particular, $x_n^* x_n$ is invertible, and hence x_n is invertible, for all sufficiently large n . At those indices put $u_n = x_n(x_n^* x_n)^{-1/2}$. Then u_n is unitary, and, once $\|x_n^* x_n - 1\| \leq \frac{1}{2}$, continuous functional calculus gives the quantitative estimate

$$\|u_n - x_n\| \leq 2 \|x_n\| \|x_n^* x_n - 1\|.$$

The sequence (x_n) is bounded, so the right-hand side tends to 0. Setting $u_n = 1$ at the finitely many remaining indices therefore gives a unitary sequence with $\|u_n - x_n\| \rightarrow 0$. $\square$

Lemma 6.2 (unitaries in the quotient). *For every sequence (d_n) , the map*

$$\kappa_{(d_n)}: \mathcal{U}_{\text{cor}}((d_n)) \longrightarrow \mathcal{U}(\mathcal{Q}), \quad [(u_n)] \longmapsto q((u_n)),$$

is a canonical group isomorphism. [Lean: GroupApproximation.ManuscriptExactWrappers.manuscriptUnitaryCoronaEquivalence]

Proof. The map is well defined, and its kernel consists exactly of the unitary sequences satisfying $\|u_n - 1\| \rightarrow 0$, so it is injective. It is surjective by Lemma 6.1: every unitary in $\mathcal{Q}$ has a unitary lift (u_n) , whose class in (2) maps to it. $\square$

7 KAZHDAN TRANSPORT AND FINITE-NORMAL CANCELLATION

The corona half of Theorem A will be deduced from a general criterion, whose data we now fix.

Definition 7.1 (marked Kazhdan pattern). Let H be a countable group. A *marked Kazhdan pattern* in H is a tuple (Λ, ι, t, c) , where Λ is a countable group with property (T), $\iota: \Lambda \rightarrow H$ is a group homomorphism — *not* assumed injective — and $t, c \in H$, subject to

(M1) $t \iota(\lambda) t^{-1} \in \iota(\Lambda)$ for every $\lambda \in \Lambda$;

(M2) $c \iota(\lambda) = \iota(\lambda) c$ for every $\lambda \in \Lambda$.

Write $d := t c t^{-1}$. The *compression-defect subgroup* of the pattern is the normal closure

$$N_{\text{comp}} = \langle\langle [d, \iota(\lambda)] : \lambda \in \Lambda \rangle\rangle_H \trianglelefteq H.$$

[Lean: GroupApproximation.ManuscriptExactWrappers.manuscriptMarkedKazhdanPattern]

Theorem 7.2 (finite-normal obstruction criterion). *Let H be a countable group with a marked Kazhdan pattern (Λ, ι, t, c) , and let $F \trianglelefteq H$ be a finite normal subgroup with $F \subseteq N_{\text{comp}}$. Then every corona representation $\Theta: H \rightarrow \mathcal{U}(\mathcal{Q})$ satisfies $\Theta(f) = 1$ for all $f \in F$.* [Lean: GroupApproximation.ManuscriptExactWrappers.manuscriptFiniteNormalObstructionCriterion]

Proof. Suppose that a corona representation Θ is nontrivial on F . Form the Reynolds projection directly in the quotient,

$$p = \frac{1}{|F|} \sum_{f \in F} \Theta(f), \quad q = 1 - p.$$

Normality of F makes p central relative to $\Theta(H)$, and nontriviality on F gives $q \neq 0$. Since q commutes with $\Theta(H)$, the formula

$$\Theta_q(g) = q\Theta(g)q$$

defines a homomorphism $H \rightarrow \mathcal{U}(q\mathcal{Q}q)$, with corner unit q . There

$$(*) \quad \sum_{f \in F} \Theta_q(f) = q \left(\sum_{f \in F} \Theta(f) \right) q = |F|qpq = 0$$

exactly.

Choose a sequence of coordinate projections (q_n) representing q ; this is obtained by applying a spectral cut to a self-adjoint lift of q . If $J = \{n : q_n \neq 0\}$, then J is infinite because $q \neq 0$. Restriction to J , followed by a relabeling of J by $\mathbb{N}$, identifies the corner with

$$\begin{aligned} q\mathcal{Q}q &\cong \prod_{n \in \mathbb{N}} q_n M_{d_n}(\mathbb{C}) q_n / \bigoplus_{n \in \mathbb{N}} q_n M_{d_n}(\mathbb{C}) q_n, \\ &\cong \prod_{n \in \mathbb{N}} M_{r_n}(\mathbb{C}) / \bigoplus_{n \in \mathbb{N}} M_{r_n}(\mathbb{C}), \quad r_n = \text{rank } q_n > 0. \end{aligned}$$

Thus zero-rank coordinates are discarded, rather than treated as matrix coordinates of size zero. Since H is countable, choose coordinate lifts of all the elements $\Theta_q(g)$. For each fixed g , its two unitary defects tend to zero; on a g -dependent cofinite tail polar correction gives $U_{g,n} \in \mathcal{U}(r_n)$, and at the finitely many remaining indices choose an arbitrary unitary. The homomorphism identity for Θ_q then makes these lifts asymptotically multiplicative. Put

$$K_2 = \{g \in H : \|U_{g,n} - 1\|_2 \rightarrow 0\}.$$

Unitary invariance and asymptotic multiplicativity make $K_2 \trianglelefteq H$. The quantitative Kazhdan fixed-space transport of Lemma 7.4 gives

$$[d, \iota(\lambda)] \in K_2 \quad (\lambda \in \Lambda).$$

Hence $N_{\text{comp}} \subseteq K_2$, and therefore $F \subseteq K_2$. Because F is finite,

$$\sum_{f \in F} U_{f,n} \rightarrow |F|1 \quad \text{in normalized Hilbert–Schmidt norm.}$$

On the other hand, the exact quotient identity $(*)$ for Θ_q says that the same sum converges to 0 in operator norm, hence also in normalized Hilbert–Schmidt norm. This contradiction proves the theorem. $\square$

Theorem A(2) is the instance $H = E$, $\Lambda = \mathcal{B}$, ι the canonical homomorphism, and $F = \{1, w\}$. The membership $w \in N_{\text{comp}}$ is the identity $w = [d, \iota(v_1)]^2$ of Lemma 8.1. The following complementary coordinate construction forms the Reynolds average at finite stages and repairs its complementary spectral corner coordinate by coordinate.

7.1 Quantitative realization of Kazhdan cancellation The remainder of this section proves the dimension-free coordinate estimates used above. Spectral cuts and polar correction are performed before taking the quotient, and the reverse leakage estimate is obtained from equality of finite coordinate ranks.

Lemma 7.3 (coordinate finite-normal corner). *Let H be a countable group, let $F \triangleleft H$ be finite, let $(d_i)_{i \in I}$ be a family of positive integers, let ω be an ultrafilter on I , and let ρ be a homomorphism from H into the operator-norm metric ultraproduct of the groups $\mathcal{U}(M_{d_i}(\mathbb{C}))$, $i \in I$. If $\rho|_F$ is nontrivial, then there are nonempty finite coordinate sets Y_n and unitaries $V_{g,n} \in \mathcal{U}(M_{Y_n}(\mathbb{C}))$ such that*

$$\|V_{g,n}V_{h,n} - V_{gh,n}\| \longrightarrow 0 \quad (g, h \in H), \quad \left\| \sum_{f \in F} V_{f,n} \right\| \longrightarrow 0.$$

[Lean: GroupApproximation.ManuscriptExactWrappers.manuscriptCoordinateFiniteNormalCorner]

Proof. Choose $f_0 \in F$ with $\rho(f_0) \neq 1$. A standard sequential extraction from the ultraproduct gives indices $i_n \in I$ and representatives $U_{g,n} \in \mathcal{U}(M_{d_{i_n}}(\mathbb{C}))$ which are asymptotically multiplicative in operator norm, together with a constant $\delta > 0$ such that

$$\|U_{f_0,n} - U_{1,n}\| \geq \delta$$

for every n . We now construct the corner model from these coordinates. Put

$$A_n = \frac{1}{|F|} \sum_{f \in F} U_{f,n}, \quad H_n = \frac{1}{2}(A_n + A_n^*).$$

Left and right translation permute F , while conjugation by any $g \in H$ permutes F by normality. Expanding the finitely many multiplicative defects therefore shows, in operator norm, that

$$A_n^2 - A_n \longrightarrow 0, \quad A_n - H_n \longrightarrow 0, \quad U_{g,n}A_nU_{g,n}^* - A_n \longrightarrow 0.$$

Let $P_n = \chi_{(1/2, \infty)}(H_n)$ and $Q_n = 1 - P_n$. Spectral rounding gives

$$\|P_n - H_n\| \leq 2 \|H_n^2 - H_n\| \longrightarrow 0,$$

so $P_n - A_n \rightarrow 0$ and $U_{g,n}P_nU_{g,n}^* - P_n \rightarrow 0$ for every fixed g . The separation of f_0 forces $Q_n \neq 0$ eventually: if $Q_n = 0$, then $P_n = 1$, and the

relations $U_{f_0,n}A_n - A_n \rightarrow 0$, $P_n - A_n \rightarrow 0$, and $U_{1,n} - 1 \rightarrow 0$ would imply $\|U_{f_0,n} - U_{1,n}\| < \delta$. Discard these finitely many indices and relabel, so that $Q_n \neq 0$ for every n .

Conjugate into an eigenbasis of H_n , choose a nonempty finite set Y_n indexing $Q_n \mathbb{C}^{d_{in}}$, and let $C_{g,n} \in M_{Y_n}(\mathbb{C})$ be the corresponding principal block of $U_{g,n}$. The preceding commutator estimate makes both off-diagonal blocks tend to zero. Hence

$$\|C_{g,n}C_{h,n} - C_{gh,n}\| \rightarrow 0, \quad \|C_{g,n}^*C_{g,n} - 1\| \rightarrow 0.$$

For each fixed g and all sufficiently large n the compressed Gram matrix is invertible; take the polar correction

$$V_{g,n} = C_{g,n}(C_{g,n}^*C_{g,n})^{-1/2}.$$

When the Gram defect is at most $1/2$, functional calculus gives the dimension-free estimate

$$\|V_{g,n} - C_{g,n}\| \leq 2\|C_{g,n}^*C_{g,n} - 1\|.$$

For each fixed g , use this polar unitary on the cofinite tail and set $V_{g,n} = 1$ at its finitely many exceptional indices. Thus polar correction is asymptotically invisible and the $V_{g,n}$ remain asymptotically multiplicative.

Finally, the Q_n -block of A_n is the normalized sum. Since $P_n - A_n \rightarrow 0$ and $Q_n P_n = 0$,

$$\left\| |F|^{-1} \sum_{f \in F} C_{f,n} \right\| \rightarrow 0.$$

Multiplying by the fixed factor $|F|$ gives $\|\sum_{f \in F} C_{f,n}\| \rightarrow 0$. There are only finitely many $f \in F$, so replacing every $C_{f,n}$ by its polar correction preserves this limit and proves the lemma. $\square$

The second ingredient is the finite-stage Kazhdan-compression calculation.

Lemma 7.4 (Kazhdan fixed-space transport). *Let (r_n) be a sequence of positive integers and let $V_{g,n} \in \mathcal{U}(r_n)$ be any operator-norm asymptotic unitary representation of H . For a marked Kazhdan pattern (Λ, ι, t, c) , with $d = tct^{-1}$, for every $\lambda \in \Lambda$ and $\varepsilon > 0$ there is N such that, for all $n \geq N$,*

$$\|V_{[d, \iota(\lambda)], n} - V_{1,n}\|_2^2 \leq \varepsilon.$$

[Lean: GroupApproximation.ManuscriptExactWrappers.manuscriptCompressionDefectsCollapse]

Proof. Apply the adjoint microstates $\text{Ad } V_{\iota(\lambda), n}$ to $K_n = L^2(M_{r_n}, \text{Tr})$ with its unnormalized Frobenius inner product. In this proof $\|\cdot\|_2$ denotes the resulting Euclidean norm, so $\|x\|_2 = r_n^{-1/2}\|x\|_2$. Choose a finite symmetric Kazhdan set $S \subseteq \Lambda$, with $1 \in S$, generating Λ , and a Kazhdan constant $0 < \kappa \leq 1$. Fix

$$1 - \frac{\kappa^2}{4|S|} < \theta < 1,$$

let H_n be the Hermitian average of the adjoint S -microstates, and put $P_n = \chi_{(\theta, \infty)}(H_n)$. Spectral capture gives, for every fixed $\lambda \in \Lambda$,

$$\|(\text{Ad } V_{\iota(\lambda), n} - 1)P_n\| \longrightarrow 0.$$

Here are the details behind this dimension-free assertion. If, along a subsequence, H_n had an eigenvalue in a fixed interval

$$[\theta, 1 - \delta] \quad (\delta > 0),$$

take corresponding unit eigenvectors and pass to their Hilbert-space ultraproduct. The operator-norm multiplicative defects vanish there, so the adjoint microstates define an actual unitary representation of Λ . The limiting vector is an eigenvector of the Hermitian S -average with eigenvalue at least θ and strictly below 1. It is therefore orthogonal to the invariant subspace. On that orthogonal complement the Kazhdan-pair inequality bounds the Hermitian average by $1 - \kappa^2/(4|S|)$, contradicting the choice of θ . Hence the spectrum of H_n above θ is eventually contained arbitrarily close to 1, uniformly in r_n .

For $P_n = \chi_{(\theta, \infty)}(H_n)$ this says $\|(H_n - 1)P_n\| \rightarrow 0$. The exact identity

$$\sum_{s \in S} ((\text{Ad } V_{\iota(s), n} - 1)P_n)^* ((\text{Ad } V_{\iota(s), n} - 1)P_n) = 2|S| P_n(1 - H_n)P_n$$

then gives the asserted displacement first for $s \in S$. The set of group elements having vanishing displacement is closed under products: expand the displacement of a product, use the two known displacements, and absorb its operator-norm multiplicative defect. Since the symmetric set S generates Λ , the conclusion follows for every fixed λ .

Put $\xi_n = \text{vec}(V_{c, n})$, so $\|\xi_n\|_2^2 = r_n$. Because c centralizes $\iota(\Lambda)$, asymptotic multiplicativity gives

$$r_n^{-1/2} \|(I - H_n)\xi_n\|_2 \longrightarrow 0.$$

The normalization is supplied by the elementary energy estimate

$$0 \leq \text{Re}\langle \xi_n, (I - H_n)\xi_n \rangle \leq \|\xi_n\|_2 \|(I - H_n)\xi_n\|_2 = o(r_n).$$

Since the spectrum of H_n on $(1 - P_n)K_n$ is at most θ , functional calculus gives

$$(1 - \theta)\|(1 - P_n)\xi_n\|_2^2 \leq \text{Re}\langle \xi_n, (I - H_n)\xi_n \rangle.$$

These two inequalities give the root-capture estimate

$$(11) \quad r_n^{-1/2} \|(1 - P_n)\xi_n\|_2 \longrightarrow 0.$$

For each $s \in S$, (M1) supplies an $s' \in \Lambda$ with $t\iota(s)t^{-1} = \iota(s')$. The multiplicative defects then give, with $T_n = \text{Ad } V_{t, n}$ and $\beta_{g, n} = \text{Ad } V_{\iota(g), n}$,

$$\|\beta_{s, n} T_n^* - T_n^* \beta_{s', n}\| \longrightarrow 0.$$

Multiplying on the right by P_n and using $\|(\beta_{s',n} - 1)P_n\| \rightarrow 0$ shows that $T_n^*P_n$ is asymptotically fixed by every $\beta_{s,n}$; symmetry of S gives the same statement for the adjoints. Averaging the two statements yields

$$\|(1 - H_n)T_n^*P_n\| \rightarrow 0.$$

The operator spectral-cut inequality

$$(1 - \theta)\|(1 - P_n)T_n^*P_n\|^2 \leq \|(1 - H_n)T_n^*P_n\|$$

therefore shows that the adjoint-conjugated spectral corner has asymptotically no leakage below the same cut. More precisely, if $R_n = T_nP_nT_n^*$, unitary invariance gives

$$\|(1 - R_n)P_n\| \rightarrow 0.$$

They have equal rank at every coordinate. The finite-dimensional projection flip bounds the reverse leakage by $\eta/\sqrt{1 - \eta^2}$ whenever the forward leakage is at most $\eta < 1$. Thus

$$\|(1 - P_n)R_n\| \rightarrow 0,$$

and hence

$$\|P_n - R_n\| \rightarrow 0.$$

This equal-rank projection argument is the finite-dimensional replacement for the stable-finiteness step one might be tempted to perform only after taking a C^* -quotient; Figure 2 displays the implication.

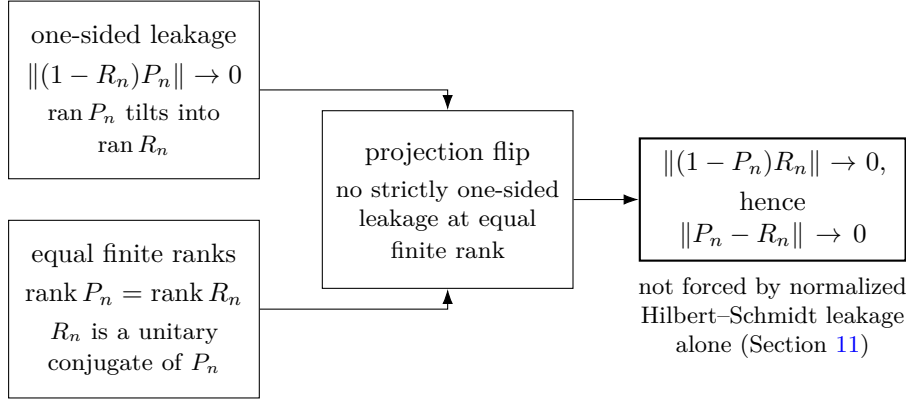


Figure 2: Equal-rank reversal. One-sided spectral leakage, together with equality of the two finite ranks, forces the reverse leakage — bounded by $\eta/\sqrt{1 - \eta^2}$ — and hence convergence of the two projections. The reversal uses equality of the finite coordinate ranks; normalized Hilbert–Schmidt-small leakage alone neither controls those ranks nor yields the reverse operator-norm estimate (Section 11).

Write $T_n = \text{Ad } V_{t,n}$. Since $R_n = T_n P_n T_n^*$, root capture and $\|P_n - R_n\| \rightarrow 0$ imply

$$r_n^{-1/2} \|(1 - P_n) T_n \xi_n\|_2 \rightarrow 0.$$

For every fixed $\lambda \in \Lambda$, spectral capture gives $\|(\text{Ad } V_{\iota(\lambda),n} - 1) P_n\| \rightarrow 0$. Splitting $T_n \xi_n$ at P_n therefore yields

$$r_n^{-1/2} \|(\text{Ad } V_{\iota(\lambda),n} - 1) T_n \xi_n\|_2 \rightarrow 0.$$

This is normalized Hilbert–Schmidt commutation of the transported lamp tct^{-1} with $\iota(\lambda)$. The operator-norm multiplicative defects identify the corresponding matrix commutator with the microstate value of $[d, \iota(\lambda)]$, and identify the identity word with $V_{1,n}$, up to an error tending to zero. This proves the lemma. $\square$

8 THE DEFECT SQUARE AND THE EXPLICIT INSTANCE

Lemma 8.1 (the marked word is a defect square). *Let H_1 be a group and $d, a_1 \in H_1$ with $d^2 = 1$. Then*

$$[d, a_1 d a_1^{-1}] = [d, a_1]^2.$$

[Lean: GroupApproximation.commutator_conjugate_eq_commutator_sq_of_sq_eq_one]

Proof. Put $b = a_1 d a_1^{-1}$, so $b^2 = a_1 d^2 a_1^{-1} = 1$; thus $b^{-1} = b$ and $d^{-1} = d$. Then $[d, a_1] = d a_1 d^{-1} a_1^{-1} = d (a_1 d a_1^{-1}) = db$, and

$$[d, a_1]^2 = (db)^2 = db db = d b d^{-1} b^{-1} = [d, b]. \quad \square$$

Proof of Theorem A. Part (1) is Proposition 4.3. For part (2), use the canonical homomorphism $\iota: \mathcal{B} \rightarrow E$ and put $a = v_1$. The tuple $(\mathcal{B}, \iota, t, c)$ is a marked Kazhdan pattern in E : Proposition 3.2 gives property (T); the six stable-letter relators give $t\iota(\mathcal{B})t^{-1} \subseteq \iota(\mathcal{B})$; and the six commutator relators give $[c, \iota(\mathcal{B})] = 1$. No injectivity of ι is required. In E we have $d^2 = (tct^{-1})^2 = tc^2t^{-1} = 1$, so Lemma 8.1 with $a_1 = \iota(v_1)$ gives

$$w = [d, \iota(v_1) d \iota(v_1)^{-1}] = [d, \iota(v_1)]^2 \in N_{\text{comp}},$$

since $[d, \iota(v_1)]$ is one of the defining generators of N_{comp} . The subgroup $F = \{1, w\} = \langle w \rangle$ is finite and normal: centrality is a defining relation and $w^2 = 1$ follows from centrality together with $c^2 = 1$, while Proposition 4.3 gives $w \neq 1$. Moreover $F \subseteq N_{\text{comp}}$. Theorem 7.2 therefore kills w after composing any $\Theta: E \rightarrow \text{U}_{\text{cor}}((d_n))$ with the canonical isomorphism $\kappa_{(d_n)}$ of Lemma 6.2. Since $\kappa_{(d_n)}$ is injective, $\Theta(w) = 1$. This is part (2).

For the final assertions of Theorem A: an MF embedding of E would be an injective corona representation, which is impossible because every corona representation sends the nontrivial element w to 1. If $C_{\text{red}}^*(E)$ were an MF algebra, an injective, possibly nonunital $*$ -homomorphism $e: C_{\text{red}}^*(E) \rightarrow \mathcal{Q}$ would give the injective unitary homomorphism $u \mapsto e(u) + (1 - e(1))$. Composing it with the canonical map $E \rightarrow \mathcal{U}(C_{\text{red}}^*(E))$, and then with $\kappa_{(d_n)}^{-1}$, would exhibit an injective corona representation of E : the map $g \mapsto \lambda_g$ is

injective. The canonical map $E \rightarrow \mathcal{U}(C_{\max}^*(E))$ is injective as well, because coordinate evaluation at the reduced left-regular representation sends u_g to λ_g . Thus an MF embedding of $C_{\max}^*(E)$ would again produce an injective corona representation of E after the same $\kappa_{(d_n)}^{-1}$ conversion. $\square$

The marked radical element. The proof shows more than non-injectivity of each individual Θ : the fixed element w lies in the kernel of *every* corona representation of E , so every corona representation factors through $E/\langle w \rangle$ (note $\langle w \rangle = \{1, w\}$ is central, hence normal). In the language of approximation radicals, w is a nontrivial element of the “MF radical” of E . This terminology is meant in analogy with Slofstra’s earlier quotients G^{fin} and G^{fa} , obtained by killing elements invisible to exact and approximate finite-dimensional representations, respectively [Slo17, Definitions 2.5–2.7]. By Theorem 7.2 the same holds in every countable group with a marked Kazhdan pattern: every finite normal subgroup of the compression-defect subgroup lies in the MF radical. Section 9 develops the radical systematically; we do not know whether $E/\langle w \rangle$ is MF, i.e. whether the MF radical of E is exactly $\{1, w\}$. No density hypothesis. No lower bound on the corner ranks r_n , or on their density in d_n , is used anywhere: Lemma 7.3 equips each nonzero complementary block with its own normalized trace. Thus all subsequent Hilbert–Schmidt estimates are relative to that block. This is what makes the operator-norm setting insensitive to small relative ranks, which block the same argument in Hilbert–Schmidt metrics (Section 11).

9 CONSEQUENCES

Throughout this section, “corona representation” and “MF” are as in Section 2.

We record the universal MF radical, the reduced-group- C^* consequence, and the failure of quotient permanence. These are direct consequences of the same literal marked element; no additional envelope or recursion-theoretic construction is used.

9.1 The MF radical

Definition 9.1. For a countable group G_1 , the *MF radical* is

$$\text{Rad}_{\text{MF}}(G_1) = \bigcap \{ \ker \Theta : \Theta \text{ a corona representation of } G_1 \},$$

a normal subgroup of G_1 . This is the operator-norm analogue of the invisible subgroups of [Slo17, Definitions 2.5–2.7] and of the metric-ultraproduct approximation framework of [NST]. [Lean: GroupApproximation.ManuscriptExactWrappers.manuscriptMFRadical]

Lemma 9.2 (portability). *For every homomorphism $\varphi: G_1 \rightarrow G_2$ of countable groups,*

$$\varphi(\text{Rad}_{\text{MF}}(G_1)) \subseteq \text{Rad}_{\text{MF}}(G_2) :$$

the MF radical is functorial. In particular, if $\varphi: E \rightarrow G_2$ satisfies $\varphi(w) \neq 1$, then G_2 is not MF. [Lean: GroupApproximation.ManuscriptExactWrappers.manuscriptRadicalPortability]

Proof. If $x \in \text{Rad}_{\text{MF}}(G_1)$ and Θ is a corona representation of G_2 , then $\Theta \circ \varphi$ is a corona representation of G_1 , so $\Theta(\varphi(x)) = 1$. For the second statement: $w \in \text{Rad}_{\text{MF}}(E)$ by Theorem A, so $1 \neq \varphi(w) \in \text{Rad}_{\text{MF}}(G_2)$, and an injective corona representation of G_2 would kill $\varphi(w)$, a contradiction. $\square$

Proposition 9.3 (the universal MF quotient). *Let G_1 be a countable group and $R = \text{Rad}_{\text{MF}}(G_1)$.*

- (1) *R is the kernel of a single corona representation of G_1 .*
- (2) *G_1/R is MF.*
- (3) *Every homomorphism from G_1 to an MF group factors through G_1/R .*

In particular G_1 is MF if and only if $R = \{1\}$. The statement is presumably folklore once the radical is defined; we include a proof for completeness and claim no novelty for it. [Lean: GroupApproximation.ManuscriptExactWrappers.manuscriptUniversalMFQuotient]

Proof. (3) If $\varphi: G_1 \rightarrow M$ with M MF and $j: M \rightarrow \text{U}_{\text{cor}}((d_n))$ is an injective corona representation, then $j \circ \varphi$ is a corona representation of G_1 , so $j(\varphi(R)) = 1$ and, by injectivity of j , $\varphi(R) = 1$.

(1) and (2). Work on $Q = G_1/R$ from the outset. If $1 \neq q \in Q$ and $g \in G_1$ represents q , then $g \notin R$. Hence some corona representation of G_1 detects g . Every corona representation kills R by definition, so this representation descends to one of Q that detects q . Thus the intersection of the kernels of all corona representations of the countable group Q is trivial.

Enumerate the nonidentity elements of Q and choose a detecting corona representation for each. Apply the standard block-diagonal local-model diagonalization on Q : at stage k , impose multiplicativity within $1/k$ on the first k group elements in the first k representations and retain a fixed positive separation for each of the first k marked elements. This gives one faithful representation of Q in a norm matrix corona and proves (2). Composing it with $G_1 \rightarrow Q$ gives a single corona representation of G_1 with kernel exactly R , proving (1). This quotient-first construction has no separate radical-killing condition to lose in a growing block maximum. For the final equivalence: if G_1 is MF, an injective corona representation forces $R = \{1\}$; if $R = \{1\}$, then (2) says $G_1 = G_1/R$ is MF. $\square$

Corollary 9.4 (exact radical from a candidate quotient). *Let $N \trianglelefteq G_1$ satisfy $N \leq \text{Rad}_{\text{MF}}(G_1)$. Every corona representation of G_1 factors uniquely through G_1/N . If in addition G_1/N is MF, then*

$$\text{Rad}_{\text{MF}}(G_1) = N.$$

[Lean: GroupApproximation.ManuscriptExactWrappers.manuscriptExactRadicalFromCandidateQuotient]

Together with Proposition 9.3, this says that proving a concrete defect subgroup invisible and its quotient MF computes both the entire MF radical and the universal MF quotient.

Proof. The containment $N \leq \ker \Theta$ gives the unique quotient factorization for every corona representation Θ . For the reverse containment in the displayed equality, compose the quotient map $G_1 \rightarrow G_1/N$ with an injective corona representation of G_1/N : its kernel is exactly N , while the MF radical lies in the kernel of every corona representation. $\square$

Corollary 9.5. *Let A be a unital C^* -algebra admitting an injective, possibly nonunital, $*$ -homomorphism into a norm matrix corona. Then the group E admits no injective homomorphism into $\mathcal{U}(A)$.* [Lean: GroupApproximation.LiteralNonMFConsequences.literal_no_faithful_corona_subalgebra_target]

Proof. Let $j: A \rightarrow Q$ be the possibly nonunital inclusion into the ambient corona and let $\phi: E \rightarrow \mathcal{U}(A)$. Complement correction gives

$$\hat{j}(u) = j(u) + 1 - j(1_A) \in \mathcal{U}(Q).$$

This is an injective homomorphism on $\mathcal{U}(A)$, so $\hat{j} \circ \phi$ is a corona representation of E . The exact literal kernel theorem sends w to the identity there; injectivity of $\hat{j}$ gives $\phi(w) = 1$. Proposition 4.3 has $w \neq 1$, so ϕ cannot be injective. $\square$

9.2 The reduced group algebra of E

Lemma 9.6. *Let G_1 be a countable discrete group.*

- (1) *The canonical tracial state τ on $C_{\text{red}}^*(G_1)$ is faithful.*
- (2) *Every unital C^* -algebra A with a faithful tracial state is stably finite: for every nonempty finite index set I , every isometry in $M_I(A)$ is a unitary. In particular this holds in $M_k(A)$ for every $k \geq 1$.* [Lean: GroupApproximation.ManuscriptExactWrappers.manuscriptFaithfulTraceAndStableFiniteness]

Proof. (1) Realize $C_{\text{red}}^*(G_1) \subseteq B(\ell^2 G_1)$ as the norm closure of the span of the left translation unitaries λ_g , and let ρ_g be the right translation unitaries, $(\rho_g \xi)(h) = \xi(hg)$; each ρ_g commutes with each $\lambda_{g'}$, hence with every element of $C_{\text{red}}^*(G_1)$. The canonical trace is $\tau(x) = \langle x\delta_e, \delta_e \rangle$. If $\tau(x^*x) = 0$ then $\|x\delta_e\| = 0$, so $x\delta_e = 0$, and then $x\delta_g = x\rho_{g^{-1}}\delta_e = \rho_{g^{-1}}x\delta_e = 0$ for every $g \in G_1$. The vectors δ_g are total in $\ell^2 G_1$, so $x = 0$.

(2) On $M_k(A)$ the functional $\tau_k([x_{ij}]) = \frac{1}{k} \sum_i \tau(x_{ii})$ is a state with the trace property $\tau_k(xy) = \frac{1}{k} \sum_{i,j} \tau(x_{ij}y_{ji}) = \tau_k(yx)$, and it is faithful: the i -th diagonal entry of x^*x is $\sum_j x_{ji}^*x_{ji}$, so $\tau_k(x^*x) = 0$ forces $\tau(x_{ji}^*x_{ji}) = 0$, hence $x_{ji} = 0$, for all i, j . If $v \in M_k(A)$ satisfies $v^*v = 1$, then $y := 1 - vv^*$ is a projection with $\tau_k(y) = \tau_k(1) - \tau_k(vv^*) = \tau_k(1) - \tau_k(v^*v) = 0$; since $y = y^*y$, faithfulness gives $y = 0$, i.e. $vv^* = 1$. $\square$

Proof of Theorem B. $C_{\text{red}}^*(E)$ is unital and separable since E is countable; by Lemma 9.6 it carries a faithful tracial state and is stably finite. It is not an MF algebra by Theorem A. $\square$

9.3 Failure of quotient permanence

Corollary 9.7 (quotients). *MF groups are not closed under quotients: the free group F_8 is MF, and the literal group E is its quotient but is not MF.*

[Lean: GroupApproximation.LiteralMFQuotientControls.manuscriptQuotientNonclosure]

Proof. Map the eight free generators onto the eight displayed generators of E . The resulting homomorphism is surjective because those generators define the literal presentation. The group F_8 is residually finite and hence MF: the regular representations of a descending family of separating finite quotients give an injective norm-corona representation. Theorem A shows that E is not MF. $\square$

10 THE CYCLIC-BASE COMPARISON

We prove Theorem 1.7 for the literal seven-relator presentation. The stable relation gives $t\langle\gamma_0\rangle t^{-1} \subseteq \langle\gamma_0\rangle$, and c centralizes this cyclic base. The same commutant argument as in Theorem 1.6 therefore maps w_{BS} to the identity in every finite-dimensional representation over every field.

For nontriviality, use the affine realization of $\text{BS}(1, 2)$ and its coset action. The root site is fixed by the cyclic base, while the two sites represented by t and $\gamma_0 t$ are distinct. Attach Clifford generators to these sites and map c to the root lamp. All seven relators hold, and

$$w_{\text{BS}} \mapsto [c_{t\langle\gamma_0\rangle}, c_{\gamma_0 t\langle\gamma_0\rangle}] = -1.$$

Thus $w_{\text{BS}} \neq 1$. Restricting the target to the subgroup generated by the three displayed images gives the asserted surjective realized quotient. This comparison isolates the precise limitation of the finite-dimensional argument: exact finite-dimensional invisibility does not itself imply that the marked word is trivial.

11 THE OPERATOR-NORM BOUNDARY

The proof is genuinely operator-norm in its present form. If multiplication is controlled only in normalized Hilbert–Schmidt norm, the off-diagonal blocks cut out by spectral projections need not be small in operator norm. The compressed maps therefore need not admit the dimension-free polar repair used in Lemma 7.3. More importantly, Hilbert–Schmidt-small leakage does not determine finite ranks and does not imply reverse operator-norm leakage in (4).

This identifies a boundary of the method, not a tracial nonapproximation theorem for E . Alekseev–Thom ask whether commutants arising in sofic approximations of Kazhdan groups can be recovered from finite-dimensional

coordinate subalgebras [AT, Open Problem 6.2]. Such a coordinate description is separate from the operator-norm theorem proved here.

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